

Tutorial 08

ENVX2001 – Applied Statistical Methods

Semester 1

Welcome to week 8!

This week's lecture introduced you to Simple- and Multiple- Linear Regression. For both of these regression techniques we defined the key steps in model fitting and how to interpret model output.

In this tutorial, we will explore key differences between simple linear regression (SLR) and multiple linear regression (MLR), and we will learn how to fit these models in R. We will also discuss how to test assumptions and evaluate model fit for both types of regression.

You will learn how to:

1. Distinguish key similarities and differences between simple linear regression (SLR) and multiple linear regression (MLR) models
2. Implement the fitting process for each type of model
3. Test assumptions and evaluate model fit of SLR and MLR models.

Key functions to use for this week?

- show individual performance plots
- show how to interpret base plots for assumptions

Use built in dataset, should have a good eg of step function (need something good at least!)

OR can use dippers dataset at end

explain that step function and AIC function use different methods to calculate, that's why AICs are different between step and AIC.

Setting up for our analysis - FIX THIS UP

Packages

This tutorial uses tidyverse, readxl, moments, psych, and performance. Install any you are missing by running the following **in the console**:

CODE

```
install.packages(c("tidyverse", "readxl", "moments", "psych", "performance"))
```

If a package is already installed, R will simply reinstall it — no harm done.

The data



Figure 1: Cute little Dipper standing on a rock. Image source: markmedcalf via Adobe Stock.

Dippers are thrush-sized birds living mainly in the upper reaches of rivers, which feed on benthic invertebrates by probing the river beds with their beaks. The dataset in this exercise contains data from a biological survey which examined the nature of the variables thought to influence the breeding of British dippers.

Twenty-two sites were included in the survey. Some of the variables have been transformed.

Data can be found in: *Dippers.xlsx*?????

The variables measured were:

- Altitude site altitude
- Hardness water hardness
- RiverSlope river-bed slope
- LogCadd the numbers of caddis fly larvae, transformed
- LogStone the numbers of stonefly larvae, transformed
- LogMay the numbers of mayfly larvae, transformed
- LogOther the numbers of all other invertebrates collected, transformed
- Br_Dens the number of breeding pairs of dippers per 10 km of river

In the analyses, the four invertebrate variables were transformed using a $\log(x + 1)$ transformation.

Analysis

Exercise 1: Backward elimination in R

CODE

```
library(readxl)
Dippers <- read_xlsx("data/mlr.xlsx" , sheet = "Dippers")
glimpse(Dippers)
```

OUTPUT

```

Rows: 22
Columns: 8
$ Altitude <dbl> 259, 198, 251, 184, 145, 145, 198, 160, 251, 159, 160, 145,...
$ Hardness <dbl> 12.20, 22.00, 26.30, 22.50, 29.50, 39.90, 42.80, 59.60, 69...
$ RiverSlope <dbl> 10.90, 14.70, 6.90, 4.60, 1.91, 5.00, 6.20, 14.30, 4.60, 3...
$ Br_Dens <dbl> 3.60, 4.30, 3.80, 3.40, 3.80, 4.50, 4.30, 5.00, 4.50, 3.40,...
$ LogCadd <dbl> 2.303, 2.890, 3.784, 4.419, 3.219, 3.932, 3.664, 4.431, 3.7...
$ LogStone <dbl> 5.242, 4.344, 5.231, 5.242, 3.829, 4.898, 4.357, 6.337, 5.4...
$ LogMay <dbl> 0.000, 3.401, 5.826, 5.749, 5.509, 5.749, 5.371, 0.000, 4.8...
$ LogOther <dbl> 1.386, 1.609, 1.386, 1.386, 1.099, 3.045, 1.386, 2.944, 2.5...

```

You may explore the data on your own (hint: look at last week's exercise on histogram and scatterplot matrices).

When ready, perform a backward elimination starting from the full model:

CODE

```
FullMod <- lm(Br_Dens ~ ., data=Dippers)
RedMod <- step(FullMod, direction = "backward")
summary(FullMod)
summary(RedMod)
AIC(FullMod, RedMod)
```

Question 1

Which model is chosen? Why?

i Checkpoint

You should now be able to ...[PUT THIS AFTER EACH OBJECTIVE]

Wrap-up

In this tutorial, we explored factorial designs and their analysis using ANOVA. We learned about main effects, interactions, and blocking effects, and how to interpret ANOVA results in the context of factorial experiments. We also visualised these effects using ggplot2 and emmeans, enhancing our understanding of the relationships between factors and response variables in experimental designs.

Example exam questions

- 1. In a 2-way factorial design with factors A (2 levels) and B (3 levels), describe how you would identify and interpret main effects and interactions using ANOVA. Provide an example of how to visualise these effects.**
2. Explain the difference between a treatment design and an experimental design. Provide examples of each and discuss how they can be combined in a factorial design with blocking.
- 3. If you conducted a factorial experiment with blocking and found a significant interaction between two factors, how would you interpret this result? Would you still consider the main effects of each factor? Justify your answer with reference to ANOVA principles.**